

VAR2 — Subterranean Thermal Reservoir Mapping

BT2024167 | Polynomial Regression Report

1. Problem Description

VAR2 predicts the Thermal Anomaly Score (y) at surveyed locations. The inputs are x_1 , East-West coordinate offset; x_2 , North-South coordinate offset; and x_3 , vertical depth offset relative to the basecamp. The assignment describes mapping this score from sampled points to support identification of planned drilling sites.

2. Dataset Description

The personalized VAR2 training dataset contains 1,000 rows with three inputs and target y ; the test dataset contains 1,000 rows with the same three inputs. All columns are numeric and there are no missing values. Training has no duplicate rows; test has 10 duplicate feature rows. Training features range from -1 to 1 . The training target ranges from -29.6088 to 39.5182 (mean 2.1822). No ID or index columns were present.

3. Methodology

The training data was divided into 80% training and 20% validation using `random_state=42`. Polynomial regression models for degrees 1 through 20 were implemented with `PolynomialFeatures(degree=d, include_bias=False)` followed by `LinearRegression`. Validation MSE and R^2 were used for model selection. The test set and its hidden targets were not used for selection. The inputs were already bounded in $[-1, 1]$, so no additional scaling was applied.

4. Polynomial Degree Selection

Validation MSE improved from degree 1 through degree 8, which had the lowest MSE (0.244339735) and highest R^2 (0.99511839). Performance worsened from degree 9 onward. From degree 15 onward, validation errors became extremely large and R^2 was strongly negative, indicating numerical instability at these high degrees. Degree 8 is selected based on the observed validation evidence.

Degree	Polynomial Features	Validation MSE	Validation R^2
1	3	36.05116576	0.2797414484
2	9	22.77686891	0.5449457938
3	19	9.138247432	0.8174289036
4	34	2.992257925	0.9402183171
5	55	1.298492164	0.974057702
6	83	0.5091494223	0.9898278123
7	119	0.3008279579	0.9939898224
8	164	0.244339735	0.9951183885
9	219	0.2516805195	0.9949717286
10	285	0.3421606027	0.9931640463
11	363	0.3873860326	0.9922604971
12	454	0.7132914843	0.9857493016
13	559	3.976373186	0.9205568882
14	679	22.38795978	0.5527157264
15	815	250696900.8	-5008618.913
16	968	50625580.13	-1011436.668
17	1139	37284503.47	-744898.1428
18	1329	35987537.57	-718986.3377
19	1539	31362686.86	-626587.4316
20	1770	29918828.49	-597740.8934

Table 1. Degree-wise validation results on the held-out training-data split.

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5. Numerical Stability and Five-Fold Cross-Validation

High-degree features became ill-conditioned. For degrees 15-20, design rank was 793 versus 815-1,770 columns; the smallest/largest singular-value ratio was about $7e-17$. Degree 20's largest coefficient was $1.19e6$ and validation MSE 29.9 million. No numerical warnings occurred, but rank deficiency and validation results indicate instability. Degrees 1-14 retained full rank.

A shuffled five-fold cross-validation check using `KFold(n_splits=5, shuffle=True, random_state=42)` was run for the three best split-validation candidates, degrees 7, 8, and 9. Degree 8 also achieved the lowest mean CV MSE and highest mean CV R^2 , supporting its selection.

Degree	CV MSE Mean	CV MSE Std	CV R^2 Mean	CV R^2 Std	Train MSE Mean
8	0.2580737336	0.03196601467	0.9940092069	0.00078407019	0.1640154865
9	0.279380426	0.03921003825	0.993453165	0.00143811936	0.1432954048
7	0.3270459067	0.0343117242	0.9923307841	0.00156010159	0.2235498434

6. Final Model and Prediction

The selected pipeline was `PolynomialFeatures(degree=8, include_bias=False)` plus `LinearRegression`. It was refit on all 1,000 VAR2 training samples and used to predict all 1,000 test rows in their original order. Predictions were saved as **BT2024167_pred_var2.csv**, with exactly one column, y , and no index column. The split-validation scores (MSE 0.244339735, R^2 0.99511839) and cross-validation scores are training-only estimates; no test MSE or R^2 is reported because the test targets are hidden.

7. Conclusion

Degree 8 was selected because it achieved the best validation MSE and R^2 among degrees 1–20 and was also supported by five-fold cross-validation. Higher-degree models showed sharply worse validation performance and rank-deficient polynomial feature matrices. The degree-8 model was retrained on all training data to create the VAR2 submission.